

Instructor: Date:

Learner: Regional Branch:

Program: Center:

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Let's Start!

Prototypes or brand new classes?



Learning Outcome:

Describe common activities at work to a visitor using short sentences in familiar spoken and written situations.
Talk about particular topics related to software programming.

Can Do Statements:

- I can explain what prototype-based programming is.
- I can talk about the first prototype-oriented programming language.
- I can identify key features related to prototype-based programming.

Ask your partner:

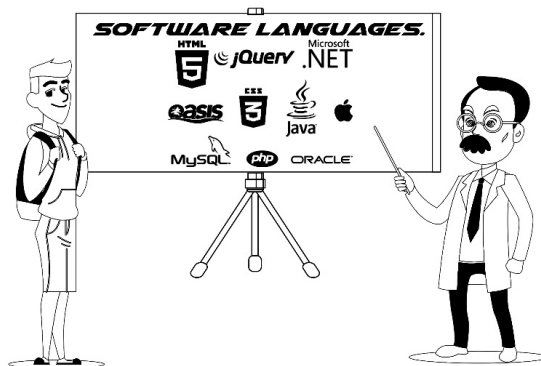
- Is it better to copy an existing object or to create it from scratch? Explain.
- Do you think inheritance is important when programming? Why?





Let's Explore!

Richard is taking a software class before taking the test



Instructions: Read the text below.

Professor: Hi, guys! Before starting, I want you to have clear some major concepts related to the programming activities we have to develop in class.

Richard: Cool! I need to know some features to differentiate those programming activities after the explanation, is that okay?

Professor: Sure! Let's start by reviewing prototype-based programming, okay?

Richard: Well, prototype-based is a style of OOP

Professor: Now, can you tell me what a prototype is and what OOP means?

Richard: Well, OOP refers to a programming language model. Now, prototype-based programming is similar to OOP but in this model, you reuse those prototypical objects for creating a new group of information.

Professor: Right, so you copy specific, generic information for reusing it into huge, new, informational lists.

Richard: Now, after you explain that, it is necessary to know the first prototype programming language.

Professor: Sure! Let's talk about Self.

Richard: Self is the first prototype-oriented programming language which looks for usual, researchable topics in object-oriented designs.

Professor: Exactly! Of course, after its creation, people wanted to develop more programming languages that supported the dynamic dispatching of their delegation.

Richard: Cool! Delegation is the correct and most effective method to find a match according to the prototypes of the different, big, round groups of objects. Delegation features allow us to find the right piece of simple data before finding a common prototype in different groups. That's why prototype-based language refers to slots.

Professor: Well Richard, I think you are ready to present the test.

Richard: I need to study the information we reviewed before taking the test, but I will hope for the best.



Let's Practice!

A Programming Style

Instructions: Select the correct answer.

1. Prototype-based is ____ ____ ____ object-oriented programming (OOP).
 - a. a style of
 - b. a method of
 - c. a technique of
2. So, you copy specific, generic information for reusing it ____ ____, informational lists.
 - a. into new, huge
 - b. into huge, new
 - c. into new. Huge
3. In the prototype-based language, you use delegation to find a match according to the prototypes of the ____ ____, groups of objects.
 - a. round, big, different
 - b. different, big, round
 - c. big, different, round



Let's Practice!

Is he talking about Self?

Instructions: Select the best options in the table.

	Self	Prototype-based Programming	Delegation
Before creating any other prototype-oriented programming model, they created Self.			



Let's Practice!

Before creating any other prototype-oriented programming model, they created Self.			
After we talk about object-oriented programming, we can talk about prototype-based programming.			
Self looks for usual, researchable topics after it finds object oriented designs.			
Before David Ungar and Randall Smith created any other language, they created Self.			



Extension Activity

Other prototype-based programming languages

Instructions: Complete the following steps. Fill in the Self-Assessment.

- Investigate about other prototype-based programming languages.
- Write down the main differences about these new languages.
- Make a presentation and share these differences with your classmates and teacher.

Self-Assessment

Yes No

- I investigate about other programming languages.
- I write information about other programming languages.
- I talk about other programming languages.

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Target Words

- **generic** (noun): original
e.g. Right, so you copy specific, generic information for reusing it into new, huge, informational lists.
- **dispatching** (verb): process of selecting the implementation of an operational method
e.g. Of course, after its creation, people wanted to develop more programming languages that supported the dynamic dispatching of their delegation.
- **slot** (noun): space for organizing information
e.g. Delegation features allow us to find the right piece of simple data before finding a common prototype in different groups. That's why prototype-based language refers to slots.
- **prototype** (noun): a model
e.g. Prototype-based programming is similar to OOP but in this model you reuse those prototypical objects for creating a new group of information.
- **reuse** (verb): to use again
e.g. Now, prototype-based programming is similar to OOP but in this model you reuse those prototypical objects for creating a new group of information.
- **delegation** (noun): assignation of tasks
e.g. Delegation is the correct and most effective method to find a match according to the prototypes of the different, big, round groups of objects.



Learning Points

Before and After as Conjunctions

You use **before** and **after** as conjunctions to join two clauses stating the order in which the actions happen.

You can use **before** and **after** between two events.

Use **before** when you want to say that an action is a requirement for another action to happen, or that an action will happen first and then a second action will take place.

Example:

*We need to have an object **before** using prototype based programming.*

In this example, it is necessary to have an object to be able to program using prototype based programming.

Use **after** to talk about an action that happens later in time in relation to another action.

Example:

*Clone an object **after** you see that you can copy its behavior to another object.*

Be careful when using a verb next to **before**, because you need to make sure it ends in -ing form.

Examples

- *Before starting, I want you to have clear some major concepts related to the programming activities.*
- *I need to know some features to differentiate those programming activities after the explanation.*
- *I need to study the information we reviewed before taking the test.*



Learning Points

Order of adjectives

When you are describing an object you can say many characteristics about it, those characteristics need to go in a specific order.

The order of the adjectives depends on the type of adjectives you use in the description. In the following chart, you can see the order depending on the type of adjectives you are using:

(Chart 1)

Remember the adjectives always go before the noun you are talking about.

Example:

*This is an **exceptional, brand-new, cutting-edge** style of OOP.*

*This is an **advanced, well-developed** object that we cloned.*

Order	Type	Example
1	Opinion	exceptional, nice, strange
2	Size	big, small, huge, tall
3	physical quality	thin, fat,
4	Shape	cylindrical, conical, square
5	Age	old, young, senior
6	Color	blue, green, black, red

Examples

- *So you copy specific, generic information for reusing it into huge, new, informational lists.*
- *Self is the first prototype-oriented programming language which looks for usual, researchable topics in object oriented designs.*
- *Delegation is the correct and most effective method to find a match according to the prototypes of the different, big, round groups of objects.*

